What produces a capturable complexity gap: one representative economy per model and group

Alexander Ober, with Claude

October 11, 2026; restructured October 15, final numbers October 19, 2026

Scope. Every economy with a protocol run (the committed record in `variants/results/economy_table.csv` plus the October 6–15, 2026 Quest campaign in `bop_runs/results`) and the reduced-form screens. The main tables hold one representative per model per group, built so that the same change is made in all three models; everything else that was run is in the appendix (Section 7). The tables are generated by `variants/campaign_tables.py` from the oracle and estimator files. Every campaign economy ran ten seeds (`kpgam` three, stopped by the oracle rule; the reduced form five per cell).

Reading the tables. Every table has the same columns. $\mathrm{SR}_{\max}$, the *linear ceiling* and the *room* are the population investor's conditional Sharpe ratios over the evaluation window: the linear ceiling is the best Sharpe a model linear in the ranks of the characteristics reaches with the true conditional moments, the room is the best nonlinear ceiling (random features, bins, quadratic) minus that linear ceiling, and *room % of linear* is the room as a percentage of the linear ceiling. That is the theoretical scope for nonlinearity, before any estimation. *Market share* is the equal-weighted market's share of $\mathrm{SR}_{\max}$. *DKKM*, *FMR* (Fama–MacBeth), *FF* (Fama–French factors), *linear* (the best of rank and level regressions, with and without the market) and *CAPM* (the estimated market line) are each method's out-of-sample Sharpe at the 360-month window. The *fair gap* is DKKM minus the best of the four classical methods, mean (sd) across seeds, and "+" counts the seeds in which it is positive. A dagger marks an oracle that predates the evaluation-window ceilings.

1 The answer

Each model has a spine: its published baseline, and one economy in which a priced state y enters firm cash flows through an interaction of two firm variables (group 3b). Every other group is one change to that spine, the same change in all three models. Only the interaction produces a fair gap. A single-type exposure to the same priced state (3a) gives room of 5 to 10% of the linear ceiling and no gap. A crash-only factor (1) gives room of 3 to 12% and no gap; a quadratic price of the model's own risk (2) gives room of 0 to 14% and no gap. A second priced state (4) raises room and gap only where the exposures are interactions: it doubles BGN's and GS21's gaps and lowers KP14's.

Economically: a capturable gap needs a firm's exposure to a priced shock to be the product of two things that different characteristics reveal, "what the firm is exposed to" (its type, or its productivity) times "how levered the firm's claim is to it" (growth options in KP14, the project portfolio in BGN, the coupon in GS21). Changing the distribution of the shock, the shape of its price, or the number of shocks does nothing unless the exposures have that structure; with it, more shocks scale the gap, and in the reduced form (Table 7) the scaling runs to a gap of 0.3 at ten factors.

group	the one change to the spine	room % of linear	fair gap
1. distribution of shocks	a crash-only cash-flow factor	3–12	none
2. price of risk	quadratic in the model's own state	0–14	none
3a. single-characteristic exposure	one type exposed to y	5–10	none
3b. interaction	three types (a product in GS21) exposed to y	16–47	+0.008 to +0.020
4. larger factor model	a second priced state	18–65	+0.008 to +0.056

The baselines every group is read against:

Table 1: Baselines. The three models as published, one type each: the floor every group is read against.

model	economy	SR_{max}	market share	linear ceiling	room	room % of linear	DKKM	FMR	FF	linear	CAPM	fair gap	+
BGN	`bgnbase` – baseline, no priced state beyond the rate	0.301	54%	0.250	+0.027	11	0.238	0.213	0.208	0.238	0.162	-0.0001 (0.006)	6/10
GS21	`gsbase` – baseline	0.293	95%	0.291	+0.001	0	0.283	0.265	0.261	0.286	0.278	-0.0028 (0.002)	2/10
KP14	`kpbase` – baseline, one type, no priced state	0.229	35%	0.221	+0.004	2	0.208	0.193	0.195	0.209	0.078	-0.0010 (0.006)	6/10

2 A different distribution of shocks: a crash-only factor

A cash-flow factor y (unit variance, half-life two years) multiplies every firm's cash flows as e^{by}, with one exposure b per model: 0.035 in BGN's project cash flows, $0.10\,\tilde{z}$ in GS21's output (the product economy's exposure without the coupon), 0.07 in KP14's project cash flows. Its Gaussian shocks carry no price. Its only priced event is a crash: each month, with probability 1%, y jumps to -3.5 standard deviations, and the kernel weights that transition by e^2 (about 7% a month risk-neutral); y then recovers at its ordinary mean reversion. Against the single-type economy of group 3a, the only change is that the Gaussian price is replaced by the crash.

Table 2: Group 1. A different distribution of shocks: a crash-only factor. Every firm's cash flows load on a factor y whose only priced shock is a rare crash (1% a month to -3.5 sd, kernel weight e^2).

model	economy	SR_{max}	market share	linear ceiling	room	room % of linear	DKKM	FMR	FF	linear	CAPM	fair gap	+
BGN	`bvyij` – crash-only factor: $e^{0.035y}$, the same crash	0.324	50%	0.278	+0.033	12	0.265	0.239	0.235	0.262	0.160	+0.0003 (0.010)	6/10
GS21	`gvyzi0j` – crash-only factor on $0.10\,\tilde{z}\,y$: y unpriced except the 1%/month jump to -3.5 (weight e^2)	0.386	82%	0.374	+0.011	3	0.358	0.342	0.334	0.363	0.316	-0.0047 (0.004)	1/10
KP14	`vyij` – crash-only factor: $e^{0.07y}$, y unpriced except a 1%/month jump to -3.5 (weight e^2)	0.336	21%	0.277	+0.012	4	0.259	0.241	0.249	0.265	0.066	-0.0057 (0.008)	2/10

Verdict. A crash changes the distribution of a shock every firm bears in the same proportion. The cross-section of premia it creates is a sort on the one firm state that scales the exposure, so the linear ceiling takes nearly all of it: room is 12% of the linear ceiling in BGN, 3% in GS21 and 4% in KP14, and the fair gap is zero or negative in all three (+0.000, −0.005, −0.006). Adding the same crash to the interaction economies instead (Appendix A1) leaves their room and gap at the control's in BGN and GS21 and halves KP14's.

3 A different price of risk: quadratic in the model's own state

The price of each model's own aggregate risk is multiplied by $1 + 0.25\,s^2$, capped at 3, where s is a standardized aggregate state: GS21's productivity x, and for KP14, whose own states are random walks, an unpriced stationary state y that no firm is exposed to. The multiplier never falls below 1 because KP14's growth-option discount turns negative once the price of x-risk drops below about 0.82 of its value. In BGN the price of the market shock σ_z is multiplied by the same function of the standardized rate, binned to seven equiprobable levels of the rate (1.00, 1.03, 1.17 and 1.68 from the

middle outward) so that project values keep a closed form; the date-k cash flow is priced at the bin of the rate one period before it pays.

Table 3: Group 2. A quadratic price of the model's own risk. The price of the baseline's aggregate risk becomes $1 + 0.25s^2$ times its value, capped at 3, s a standardized aggregate state.

model	economy	SR_{max}	market share	linear ceiling	room	room % of linear	DKKM	FMR	FF	linear	CAPM	fair gap	+
BGN	`bq25` – $\sigma_z \times \min(1 + 0.25\,s^2, 3)$, s the standardized rate, 7 bins	0.411	36%	0.346	+0.049	15	0.339	0.303	0.293	0.335	0.149	+0.0034 (0.015)	4/10
GS21	`gq25` – $\gamma_x \times \min(1 + 0.25\,\tilde{x}^2, 3)$	0.345	95%	0.343	+0.002	0	0.334	0.322	0.313	0.338	0.326	-0.0038 (0.002)	0/10
KP14	`kpq` – $\gamma_x \times \min(1 + 0.25\,y^2, 3)$, y unpriced	0.260	53%	0.251	+0.003	1	0.240	0.226	0.225	0.242	0.137	-0.0021 (0.004)	4/10

Verdict. A price of risk that moves with the aggregate state rotates the premium across states and leaves it linear in the ranks within one. The quadratic price gives room under 1% of the linear ceiling in GS21 and KP14, and 14% in BGN, where the rate is itself a priced state so a price that rises with its distance from the mean rotates the premium with it; the fair gap is zero in all three (+0.003, −0.004, −0.002). The regime prices tried earlier (appendix) do the same, and BGN's fast regime gave the group's one small positive number (+0.008), which a ranks-times-state linear benchmark takes.

4 Different exposures to aggregate risk

4.1 Exposure that a single characteristic captures

The priced state y is added with one type: every firm's cash flows load on it with the same exponent (the group 1 exposures, now with y's Gaussian shocks priced at 2.5). A firm's exposure to y then varies across firms only through one firm state: the mix of assets in place and growth option in BGN, productivity in GS21 (the product economy's exposure without the coupon), the growth-option share in KP14.

Table 4: Group 3a. Exposure captured by a single characteristic. A priced state y with one type: a firm's exposure varies only through one firm state (project mix, operating leverage, growth-option share).

model	economy	SR_{max}	market share	linear ceiling	room	room % of linear	DKKM	FMR	FF	linear	CAPM	fair gap	+
BGN	`bvy1ch` – one type, project cash flows $\times e^{0.035y}$	0.336	54%	0.297	+0.029	10	0.283	0.252	0.255	0.281	0.181	+0.0017 (0.007)	7/10
GS21	`gvyz10` – one type, output $\times e^{0.10\,\tilde{z}\,y}$: the product economy without the coupon	0.854	50%	0.800	+0.043	5	0.793	0.806	0.743	0.809	0.421	-0.0170 (0.010)	1/10
KP14	`vy1` – one type, cash flows $\times e^{0.07y}$	0.466	12%	0.369	+0.019	5	0.355	0.335	0.340	0.360	0.047	-0.0050 (0.005)	1/10

Verdict. A monotone function of one firm state, however nonlinear, is linear in the rank of the characteristic that reveals it. Room is 10% of the linear ceiling in BGN and 5% in GS21 and KP14, and the fair gap is zero or negative in all three (+0.002, −0.017, −0.005); in GS21 the productivity exposure alone carries a large premium (SR_{max} 0.85) that the linear ceiling captures almost entirely. The same holds for every other single-characteristic wiring tried (Appendix A3), including ones with far more room: room is necessary, not sufficient.

4.2 Exposure as an interaction of several characteristics

The same y with heterogeneity in what is exposed to it: three types of firms with exposures 0.01 / 0.035 / 0.07 (BGN) and 0.02 / 0.07 / 0.14 (KP14), level-compensated so that value at $y = 0$ does not reveal the type; and in GS21, where types never produced room (Appendix A3), the product of standardized productivity and the standardized coupon written into the production function with coefficient 0.10. Exposure is then a product: type × project mix (BGN), type × growth-option share (KP14), productivity × leverage (GS21), each factor revealed by a different characteristic.

Table 5: Group 3b. Exposure as an interaction of characteristics. The same y with three types (BGN, KP14) or a product written into production (GS21): exposure = type or productivity × how levered the claim is.

model	economy	SR_{max}	market share	linear ceiling	room	room % of linear	DKKM	FMR	FF	linear	CAPM	fair gap	+
BGN	`bvych0` – y as a 21-node chain, three types (control)	0.576	37%	0.341	+0.055	16	0.334	0.285	0.292	0.326	0.216	+0.0076 (0.009)	8/10
GS21	`gvyp10` – product exposure $\tilde{z}\tilde{b}$ at 0.10 (control)	0.650	8%	0.447	+0.194	47	0.550	0.189	0.333	0.531	0.018	+0.0196 (0.022)	9/10
KP14	`vyg25` – three types on y, price 2.5 (control)	0.495	11%	0.346	+0.082	24	0.350	0.292	0.320	0.335	0.044	+0.0148 (0.009)	10/10

Verdict. Every interaction economy has a positive mean gap with most seeds positive, at room of 16 to 47% of the linear ceiling. The premium is multiplicatively separable in the ranks, $f(A)\,g(B)$; a linear model in the ranks fits only additive functions, random features span products, and DKKM recovers 15 to 30% of the room as a gap. Going from 3a to 3b changes nothing but the second dimension of heterogeneity, and that is what creates the gap. The GS21 product ladder (Appendix A4) shows the room hump-shaped in the product's size and the gap rising to +0.08.

5 Larger factor models: two priced states

A second independent priced state, with the total price held at the one-state value ($2.5/\sqrt{2}$ on each), and three types whose loading vectors span the two states in the pattern $(1,0)$, $(0,1)$, $(1,1)/\sqrt{2}$, at the 3b exposure sizes. In GS21 each type carries the 0.10 product exposure to its state.

Table 6: Group 4. Larger factor models: two priced states. Two independent priced states, three types with loading vectors (1,0), (0,1), $(1,1)/\sqrt{2}$, total price held at 2.5.

model	economy	SR_{max}	market share	linear ceiling	room	room % of linear	DKKM	FMR	FF	linear	CAPM	fair gap	+
BGN	`bvyk2` – two priced chains, three loading vectors	0.639	36%	0.393	+0.071	18	0.394	0.332	0.353	0.379	0.228	+0.0152 (0.011)	10/10
GS21	`gvypk3` – two priced chains, three product types	0.612	8%	0.349	+0.226	65	0.465	0.245	0.306	0.407	0.050	+0.0563 (0.053)	19/20
KP14	`vym2t3` – two priced states, three loading vectors	0.475	12%	0.326	+0.090	28	0.317	0.280	0.301	0.308	0.049	+0.0075 (0.011)	8/10

Verdict. More factors are a lever only where the exposures are interactions. In BGN the second state raises room by a quarter and doubles the gap (+0.015, ten of ten seeds). In GS21 it raises the product economy's gap from +0.020 to +0.056 (nineteen of twenty seeds), as the two-type version did (appendix). In KP14, where each type's exposure is a sort within the state, a second state raises room by a sixth and lowers the gap from +0.014 to +0.008, as three states did (Appendix A5). The reduced form says the same with the structure controlled:

Table 7: Reduced form: priced factors on observed characteristics against products of latent states. Five seeds per cell, the total price of risk fixed across K (per-factor price $\propto 1/\sqrt{K}$); K = number of priced factors; L = number of latent states, products of their pairs.

configuration	SR_{max}	market share	linear ceiling	room	room % of linear	DKKM	FMR	FF	linear	CAPM	fair gap	+
kfh_lin_1 – K=1 factor on observed characteristics, linear loadings	0.598	8	0.598	+0.000	0	0.590	0.584	0.581	0.595	0.041	-0.0067 (0.009)	0/5
kfh_lin_4 – K=4	0.592	8	0.588	+0.003	0	0.555	0.578	0.567	0.575	0.041	-0.0237 (0.017)	0/5
kfh_lin_16 – K=16	0.385	13	0.383	+0.001	0	0.352	0.357	0.353	0.365	0.041	-0.0128 (0.025)	2/5
kfh_tanh_1 – K=1, bounded (tanh) loadings on observed characteristics	0.592	8	0.592	+0.001	0	0.579	0.578	0.570	0.588	0.041	-0.0105 (0.012)	0/5
kfh_tanh_4 – K=4	0.584	9	0.574	+0.007	1	0.541	0.564	0.544	0.560	0.041	-0.0232 (0.018)	0/5
kfh_tanh_16 – K=16	0.469	11	0.378	+0.044	12	0.346	0.357	0.345	0.359	0.041	-0.0152 (0.025)	1/5
kfh_sq_1 – K=1, pure curvature (squares) of observed characteristics	0.600	8	0.080	+0.519	647	0.582	0.032	0.046	0.059	0.041	+0.5161 (0.020)	5/5
kfh_sq_4 – K=4	0.601	8	0.097	+0.501	515	0.573	0.042	0.458	0.074	0.041	+0.1155 (0.011)	5/5
kfh_sq_16 – K=16	0.601	8	0.076	+0.523	687	0.571	0.036	0.429	0.052	0.041	+0.1429 (0.022)	5/5
kfo_sq_1 – K=1, squares, directions orthogonal to size	0.600	8	0.080	+0.519	648	0.582	0.032	0.048	0.059	0.041	+0.5163 (0.020)	5/5
kfo_sq_4 – K=4	0.601	8	0.092	+0.507	552	0.575	0.058	0.061	0.081	0.041	+0.4833 (0.043)	5/5
kfo_sq_16 – K=16	0.602	8	0.079	+0.520	656	0.575	0.015	0.031	0.053	0.041	+0.5030 (0.053)	5/5
lfh_L5_K1_n0 – K=1 product of latent states, no observation noise	0.598	8	0.208	+0.389	187	0.570	0.156	0.539	0.175	0.041	+0.0311 (0.017)	5/5
lfh_L5_K2_n0 – K=2	0.598	8	0.144	+0.453	314	0.564	0.111	0.520	0.115	0.041	+0.0443 (0.015)	5/5
lfh_L5_K3_n0 – K=3	0.598	8	0.106	+0.490	464	0.566	0.059	0.507	0.084	0.041	+0.0589 (0.015)	5/5
lfh_L5_K5_n0 – K=5	0.598	8	0.092	+0.502	549	0.543	0.048	0.423	0.064	0.041	+0.1193 (0.012)	5/5
lfh_L5_K10_n0 – K=10	0.598	8	0.089	+0.503	568	0.507	0.047	0.250	0.067	0.041	+0.2572 (0.014)	5/5
lfh_L5_K1_n1 – K=1 product of latent states, observation noise 1.0	0.598	8	0.167	+0.416	249	0.526	0.096	0.433	0.137	0.041	+0.0930 (0.055)	5/5
lfh_L5_K2_n1 – K=2	0.598	8	0.111	+0.469	423	0.523	0.072	0.420	0.082	0.041	+0.1025 (0.034)	5/5
lfh_L5_K3_n1 – K=3	0.598	8	0.091	+0.486	537	0.528	0.057	0.391	0.073	0.041	+0.1371 (0.030)	5/5
lfh_L5_K5_n1 – K=5	0.598	8	0.075	+0.496	664	0.496	0.038	0.309	0.060	0.041	+0.1866 (0.016)	5/5
lfh_L5_K10_n1 – K=10	0.598	8	0.079	+0.479	607	0.456	0.052	0.171	0.077	0.041	+0.2854 (0.026)	5/5

Factors that load on single observed characteristics, linear or bounded, leave no room at any K and a

negative gap; pure curvature of an observed characteristic gives room of about five times the linear ceiling and a gap of about +0.5 at every K once the loading directions are orthogonal to size (the fall with K in the unconstrained rows is Fama–French's value-weighted legs picking up size squared, since market value is exp(size)); factors that load on products of partly hidden states give room and a gap that grow monotonically with K and with the observation noise that hides the states.

6 What each economy changes

Notation. BGN (Berk, Green and Naik 1999): a firm is a portfolio of live projects; project j has systematic risk β_j, drawn from an exponential distribution, and lognormal cash flows C_j; it survives each month with probability π; a new project arrives each month and is taken if its NPV is positive; the pricing kernel prices the market shock at σ_z and the Vasicek short rate r_t. GS21 (Gomes and Schmid 2021): output is $e^{x_t+z_{it}}k_{it}$ with aggregate productivity x_t and idiosyncratic productivity z_{it} both AR(1); firms choose the coupon b_{it}, invest lumpily at a random cost, and may default; the kernel prices x-shocks at a constant $\gamma_x = 0.5$. KP14 (Kogan and Papanikolaou 2014): firm value is assets in place plus growth options; project cash flows are $x_t u_{it}\varepsilon_{jt}K_j^{\alpha}$, new projects arrive at the firm-specific rate λ_{it}, and the kernel prices disembodied productivity x at γ_x and embodied technology z at γ_z.

The added priced state y. Every economy with a y adds one stationary state with unit stationary variance and mean reversion $\kappa_y = 0.35$ per year, $y_{t+1} = \rho y_t + \sqrt{1-\rho^2}\,\eta_{t+1}$ with $\rho = e^{-\kappa_y/12}$ (an Ornstein–Uhlenbeck process in KP14, a Tauchen chain in GS21 and in BGN's chain economies). It is priced: the kernel is multiplied by $e^{-\lambda_y\eta_{t+1}-\lambda_y^2/2}$ with $\lambda_y = \gamma_v\sqrt{1/12}$, and $\gamma_v = 2.5$ unless stated. "Types" are equal shares of firms differing only in their exposure b_f to y. "Level-compensated" means type f's cash flows are also scaled by a constant $\theta_f = (V_0/V_f)^{1.2}$, the ratio of a unit project's value for type 0 and type f at $y = 0$, so that value at $y = 0$ does not reveal the type (KP14's ω device).

- **Baselines**
  - `bgnbase`: BGN as above, one type. Expected returns come from the value-weighted β_j of assets in place and from the growth option's exposure to r_t.
  - `gsbase`: GS21 as above, Table I calibration, one type.
  - `kpbase`: KP14 as above, one type, no y.
- **Group 1. A crash-only factor.** The 3a economy with the Gaussian price of y set to zero ($\gamma_v = 0$) and a crash.
  - `bvy1j`, `gvyz10j`, `vy1j`: `bvy1ch`, `gvyz10` and `vy1` with $\gamma_v = 0$ and, each month with probability 1% whatever y_t, a jump of y_{t+1} to -3.5 (the lowest node), weighted e^2 by the kernel (a risk-neutral probability of about 7% a month before renormalization). In KP14 the jump is a Poisson event at intensity 0.12 a year under P and $0.12\,e^2$ under Q.
- **Group 2. A quadratic price of the model's own risk.**
  - `bq25`: `bgnbase` with $\sigma_z \to \sigma_z\, g(r_t)$, $g(r) = \min(1 + 0.25\,((r-\bar{r})/\mathrm{sd}_r)^2, 3)$ binned to seven equiprobable bins of the stationary rate ($g = 1.00$, 1.03, 1.17, 1.68 from the middle outward); the date-k cash flow is priced at the bin of r_{t+k-1}, through seven bases $D_j(r)$ in place of the regime's two.
  - `gq25`: `gsbase` with $\gamma_x \to \gamma_x(1 + 0.25\,\tilde{x}^2)$ capped at 1.5, $\tilde{x} = x/\mathrm{sd}(x)$.

- `kpq`: `kpbase` with $\gamma_x \to \gamma_x \min(1 + 0.25\,y^2, 3)$, y the unpriced stationary state of `kpgam` ($\kappa = 0.35$, unit variance); no firm is exposed to y.

- **Group 3a. One type exposed to y.**
  - `bvy1ch`: `bgnbase` plus y (21-node chain), one type; project cash flows $C_j e^{0.035\,y_t}$.
  - `gvyz10`: `gsbase` plus y (9-node chain over ± 3.5 sd), one type; output $e^{x_t + z_{it} + 0.10\,\tilde{z}_{it}\,y_t} k_{it}$, $\tilde{z} = z/\mathrm{sd}(z)$: the product economy's exposure without the coupon.
  - `vy1`: `kpbase` plus y, one type; project cash flows $e^{0.07\,y_t} x_t u_{it} \varepsilon_{jt} K_j^{\alpha}$.

- **Group 3b. Interactions.**
  - `bvych0`: `bvy1ch` with three types, $b_f \in \{0.01, 0.035, 0.07\}$, level-compensated ($\theta_f = (V_0/V_f)^{1.2}$). Exposure is b_f times the firm's mix of assets in place and growth option.
  - `gvyp10`: `gsbase` plus y (9-node chain), one type, output $e^{x_t + z_{it} + 0.10\,\tilde{z}_{it}\tilde{b}_{it}\,y_t} k_{it}$, with $\tilde{z} = z/\mathrm{sd}(z)$ and $\tilde{b} = (b - 0.5)/0.29$ the standardized coupon. Exposure is productivity $\times$ leverage, revealed by profitability and leverage.
  - `vyg25`: `vy1` with three types, $b_f \in \{0.02, 0.07, 0.14\}$, level-compensated. Growth options are options on the $e^{b_f y}$ cash flows, so a firm's exposure is b_f times the elasticity of its value, which rises with its growth-option share: type $\times$ growth-option share, revealed by size and book-to-market.

- **Group 4. Two priced states.** Total price fixed, $\gamma_v = 2.5/\sqrt{2}$ on each.
  - `bvyk2`: `bvych0` with a second independent 21-node chain $y^{(2)}$; project cash flows

$$C_j\, e^{b_f y_t + b_f^{(2)} y_t^{(2)}}, \qquad (b_f, b_f^{(2)}) \in \{(0.01, 0),\ (0, 0.035),\ (0.049, 0.049)\}.$$

  - `gvypk3`: `gvyp10` with a second independent chain (5 nodes each, 25 joint) and three types of equal share, loading $0.10\,\tilde{z}\tilde{b}$ on y, on $y^{(2)}$, and $0.071\,\tilde{z}\tilde{b}$ on both.
  - `vym2t3`: `vyg25` with $y \in \mathbb{R}^2$, two independent OU states; type loading vectors $(0.02, 0)$, $(0, 0.07)$, $(0.099, 0.099)$ on the two.

Reduced form (Table 7)

Each economy (`variants/reduced/rf_model.py`) is a simulated panel whose true conditional means and covariances are known in closed form, so the same oracle ceilings and the same estimators run on it unchanged ($N = 500$ firms, $T = 500$ months after a 100-month burn-in, 360-month estimation window, five seeds per cell).

- **Characteristics.** Each firm has five latent AR(1) processes, one per characteristic (size, book-to-market, asset growth, profitability, momentum), with monthly persistence 0.99, 0.98, 0.90, 0.95 and 0.70 and unit variance. There are no exposure types in these ladders.

- **Factors and returns.** Excess returns follow an exact K-factor model, $r_{t+1} = B_t\lambda + B_t f_{t+1} + \varepsilon_{t+1}$, with Gaussian factor shocks of sd 0.04, idiosyncratic noise of sd 0.10, and constant prices of risk. The true premium is $\mu_t = B_t\lambda$ and the true covariance is $B_t \Omega B_t' + 0.01\, I$.

- **The market.** Every economy has a market factor whose loading is $1 + a'\tilde{X}_{it}$, linear in the standardized characteristics (weight 0.3 on book-to-market, profitability and momentum). This is a premium any linear method sees.

- **Extra priced factors.** On top of the market, K priced factors are added. The two ladders differ only in how firms load on them.
  - *Loadings on observed characteristics* (rows `kfh_`). Factor k loads on $h(w_k' \tilde{X}_{it})$, where w_k is a fixed random direction over the five standardized characteristics and h is linear, bounded (tanh), or a square (pure curvature). This is GS21's structure: exposure is a function of states the characteristics reveal directly.
  - *Loadings on products of latent states* (rows `lfh_`). The first L of the five latent processes are the firm's true states z_{it}, and the observed characteristics are those states plus persistent observation noise of sd 0 or 1.0. Factor k loads on the cross-sectionally demeaned product $z_{ia} z_{ib}$ of a pair of true states, cycling through the $L(L-1)/2$ pairs; with $L = 5$ there are ten pairs, hence K up to 10. This is the KP14 and BGN structure: a type times a within-firm state, each only partly visible in the characteristics.
- **Fixed total price.** Each extra factor's price is set so its Sharpe is $0.6/\sqrt{K}$, so the total price of risk, and $\mathrm{SR}_{\max}$ in the latent ladder, stay near 0.6 at every K. In the linear and bounded ladders $\mathrm{SR}_{\max}$ falls at $K = 16$ because sixteen directions over five characteristics are collinear and their loadings partly cancel.

The ladder therefore changes one thing at a time: the number of priced factors K, and whether their loadings are functions of what the characteristics show or products of what they only partly reveal.

7 Additional economies

Everything else that was run. The table has the main tables' columns; the bullets describe each economy against its parent (notation as in Section 6). Three groups of them answer side questions: the crash added to the interaction economies instead of to a single-type economy (`bvyj25`, `gvyp10j`, `vyj25`), the GS21 product ladder (`gvyp05` to `gvyp20`, `gvyp10x`), and the earlier single-characteristic and price-of-risk wirings that the harmonized cells replace.

Table 8: Additional economies. Everything else that was run, by model; each is described in the appendix against its parent.

model	economy	$\mathrm{SR}_{\max}$	market share	linear ceiling	room	room % of linear	DKKM	FMR	FF	linear	CAPM	fair gap	+
BGN	`g0235f` – regime price ×0.2 / ×3.5, spells 4× faster	0.237	62%	0.163	+0.045	28	0.169	0.141	0.131	0.161	0.146	+0.0079 (0.005)	10/10
BGN	`g0235df` – stress 2/3 of the time, 4× faster	0.489	17%	0.432	+0.036	9	0.423	0.392	0.351	0.428	0.077	-0.0051 (0.007)	2/10
BGN	`bvy25` – three types on y, level-compensated (1.2)	0.601	41%	0.398	+0.062	16	0.399	0.344	0.367	0.384	0.248	+0.0148 (0.016)	9/10
BGN	`bvych41` – bvych0 on a 41-node chain	0.565	38%	0.336	+0.052	15	0.328	0.279	0.291	0.320	0.215	+0.0086 (0.010)	8/10
BGN	`bvyc0` – the same without compensation	0.475	41%	0.395	+0.066	18	0.386	0.321	0.347	0.376	0.195	+0.0104 (0.011)	9/10
BGN	`bvyj25` – the chain with a priced crash (1%/month to $-y_{\max}$, priced e^2)	0.536	38%	0.315	+0.054	17	0.307	0.261	0.269	0.301	0.202	+0.0066 (0.007)	9/10
BGN	`bvysg` – signed exposure ladder (-0.07 / 0 / $+0.07$)	0.504	28%	0.361	+0.120	35	0.318	0.305	0.273	0.324	0.141	-0.0199 (0.032)	2/10
GS21	`g28` – countercyclical $\gamma(x)$, clipped	0.314	95%	0.312	+0.001	0	0.303	0.254	0.256	0.305	0.298	-0.0024 (0.002)	0/10
GS21	`bx7` – two-state regime on γ, five types on x	0.327	94%	0.317	+0.007	2	0.309	0.282	0.271	0.311	0.306	-0.0017 (0.002)	0/10
GS21	`gx7` – five types on productivity x	0.485	98%	0.474	+0.008	2	0.473	0.393	0.383	0.473	0.473	+0.0003 (0.001)	5/10
GS21	`gvy1` – one type, output $\times e^{0.05y}$	0.636	95%	0.633	+0.002	0	0.625	0.613	0.601	0.629	0.602	-0.0036 (0.003)	1/10
GS21	`gvy1j` – crash-only factor: $e^{0.05y}$, the same crash	0.379	96%	0.377	+0.001	0	0.372	0.353	0.343	0.374	0.363	-0.0027 (0.001)	0/10
GS21	`gvy25` – production-level exposure to y, three types	0.725	86%	0.709	+0.014	2	0.686	0.681	0.670	0.690	0.625	-0.0041 (0.004)	2/10
GS21	`gvys25` – signed fixed-cost exposure to y	0.619	69%	0.602	+0.014	2	0.496	0.490	0.426	0.499	0.420	-0.0032 (0.005)	2/6
GS21	`gvyp05` – product $z\bar{b}$ at 0.05	0.425	60%	0.284	+0.128	45	0.296	0.255	0.206	0.277	0.254	+0.0194 (0.010)	10/10
GS21	`gvyp075` – product at 0.075	0.561	28%	0.261	+0.287	111	0.428	0.172	0.188	0.351	0.149	+0.0765 (0.030)	10/10
GS21	`gvyp15` – product at 0.15	0.744	-25%	0.616	+0.119	22	0.661	0.327	0.515	0.583	0.192	+0.0780 (0.040)	10/10
GS21	`gvyp20` – product at 0.20	0.778	-51%	0.729	+0.039	6	0.701	0.452	0.643	0.680	0.407	+0.0215 (0.024)	9/10
GS21	`gvyp10x` – product at 0.10 plus a curvature loading on x	0.589	17%	0.365	+0.215	63	0.483	0.157	0.258	0.435	0.054	+0.0486 (0.022)	10/10
GS21	`gvyp10j` – the product exposure with a priced crash	0.537	21%	0.328	+0.197	62	0.440	0.210	0.240	0.409	0.072	+0.0301 (0.020)	10/10
GS21	`gvypk2` – two priced chains, two product types	0.478	27%	0.226	+0.212	103	0.305	0.143	0.156	0.254	0.105	+0.0510 (0.056)	9/10
KP14	`kpgam` – $\gamma_x \times$ logistic(y), 1 to 3	0.390	65%	0.373	+0.005	1	0.358	0.348	0.340	0.363	0.254	-0.0053 (0.004)	1/10
KP14	`vyx` – three types on y, price 1.8	0.421	14%	0.321	+0.055	17	0.313	0.284	0.302	0.310	0.054	+0.0009 (0.009)	7/10
KP14	`vyj25` – the same with a priced crash in y	0.554	10%	0.323	+0.104	33	0.315	0.270	0.292	0.307	0.045	+0.0080 (0.011)	8/10
KP14	`vym3t3` – three priced states, three types	0.443	14%	0.285	+0.087	30	0.279	0.246	0.266	0.273	0.061	+0.0036 (0.015)	6/10
KP14	`vym3` – three priced states, twenty rank-3 types	0.418	15%	0.298	+0.056	19	0.286	0.259	0.278	0.287	0.059	-0.0032 (0.006)	5/10

BGN

- `g0235f`: `bgnbase` with $\sigma_z \to g_s \sigma_z$, $s_t \in \{\text{calm}, \text{stress}\}$ a Markov chain, $g = 0.2$ calm and 3.5 stress; monthly switch probabilities 1/12 (calm to stress) and 1/6 (stress to calm), so stress is one third of months with a mean spell of 6 months.

- `g0235df`: the same g with switch probabilities 1/6 and 1/12: stress two thirds of months, mean spell 12 months.
- `bvy25`: `bgnbase` plus y; type f's project cash flows are $\theta_f C_j e^{b_f y_t}$, $b_f \in \{0.01, 0.035, 0.07\}$, level-compensated. Exposure is b_f times the firm's mix of assets in place and growth option.
- `bvych41`: `bvych0` with y on a 41-node chain instead of 21.
- `bvyc0`: `bvy25` with $\theta_f = 1$, so the type is visible in value.
- `bvyj25`: `bvych0` (group 3b) plus a crash: each month, with probability 1% whatever y_t, y_{t+1} goes to the lowest node; otherwise the Tauchen transition. The kernel weights the crash transition by e^2, so its risk-neutral probability is about 7.4% a month before renormalization.
- `bvysg`: `bgnbase` plus y; type f's project cash flows are $C_j e^{b_f y_t}$, $b_f \in \{-0.07, 0, +0.07\}$, not level-compensated.

GS21

- `g28`: `gsbase` with $\gamma_x \to \gamma(x_t) = \text{clip}\big(0.5 - 0.28\, x_t/\text{sd}(x),\ 0.05,\ 1.0\big)$, countercyclical.
- `bx7`: `gsbase` with $\gamma_x \to g_s \gamma_x$, $g \in \{0.6, 3.0\}$ on a two-state Markov regime, and five types with output $e^{b_f x_t + z_{it}} k_{it}$, $b_f \in \{1, 2.5, 4, 5.5, 7\}$.
- `gx7`: `gsbase` with five types, output $e^{b_f x_t + z_{it}} k_{it}$, $b_f \in \{1, 2.5, 4, 5.5, 7\}$, and $\gamma(x_t) = \text{clip}\big(0.5 - 0.4\, x_t/\text{sd}(x),\ 0.05,\ 1.5\big)$. A firm's exposure to the one priced shock is its type, which size and profitability reveal.
- `gvy1`: `gsbase` plus y (9-node chain over ± 2.5 sd), one type; output $e^{x_t + z_{it} + 0.05\, y_t} k_{it}$, a level exposure.
- `gvy1j`: `gvy1` with $\gamma_v = 0$ and the crash, on a chain over ± 3.5 sd.
- `gvy25`: `gsbase` plus y (9-node chain); output $e^{x_t + z_{it} + b_f y_t + a_f} k_{it}$, $b_f \in \{0.015, 0.05, 0.10\}$, with a_f the level compensation.
- `gvys25`: `gsbase` plus y, entering through an operating cost: $0.2\, e^{c_f y_t}$ per unit capital is subtracted from output, $c_f \in \{+0.4, 0, -0.4\}$; output itself does not load on y.
- `gvyp05, gvyp075, gvyp15, gvyp20`: `gvyp10` with the coefficient 0.10 replaced by 0.05, 0.075, 0.15 and 0.20.
- `gvyp10x`: `gvyp10` with output $e^{(1+(\tilde{z}_{it}^2 - 1)/\sqrt{2})\, x_t + z_{it} + 0.10\, \tilde{z}_{it} \tilde{b}_{it}\, y_t} k_{it}$: the loading on x is also nonlinear in productivity.
- `gvyp10j`: `gvyp10` (group 3b) plus the same priced crash on its 9-node chain.
- `gvypk2`: `gvyp10` with a second independent chain (5 nodes each, 25 joint); two types of equal share with output $e^{x_t + z_{it} + 0.10\, \tilde{z}_{it} \tilde{b}_{it}\, y_t} k_{it}$ and $e^{x_t + z_{it} + 0.10\, \tilde{z}_{it} \tilde{b}_{it}\, y_t^{(2)}} k_{it}$.

KP14

- `kpgam`: `kpbase` with $\gamma_x \to \gamma_x\, g(y_t)$, $g(y) = 1 + 2/(1 + e^{2y})$, rising from 1 when y is high to 3 when it is low; y is unpriced ($\gamma_v = 0$) and no firm is exposed to it.

- `vyx`: `kpbase` plus y with $\gamma_v = 1.8$; type f's project cash flows are $\theta_f e^{b_f y_t} x_t u_{it} \varepsilon_{jt} K_j^{\alpha}$, $b_f \in \{0.02, 0.07, 0.14\}$, level-compensated.

- `vyj25`: `vyg25` (group 3b) plus a Poisson jump of y to $-y_{\max} = -3.5$ at intensity 0.12 a year (1% a month) under P and $0.12\, e^2 = 0.89$ a year under Q.

- `vym3t3`: `vyg25` with $y \in \mathbb{R}^3$, three independent OU states; type f's cash flows load $e^{b_f' y_t}$ with three vectors $b_f \in \mathbb{R}^3$ spanning the three states.

- `vym3`: the same three states with twenty types, $b_f \in \mathbb{R}^3$ of rank 3.

What they add to the main results:

- **The chain's resolution is not what separates `bvych0` from `bvy25`**: on 41 nodes the gap is +0.009 against +0.008 on 21, and the difference from the Gaussian economy's +0.015 (sd 0.016) is within noise.

- **A level exposure of GS21 output to** y (`gvy1`, `gvy1j`) has no room at all, as every other level wiring in GS21; the productivity exposure (`gvyz10`) has 5% and the product with the coupon 43%.

- **The crash on an interaction economy** leaves room and gap at the control's in BGN (+0.007 against +0.008) and GS21 (+0.030 against +0.020, within noise) and halves KP14's (+0.008 against +0.014): the crash months are a state the ranks-times-state linear benchmark reads as well as DKKM does.

- **The GS21 product ladder**: room is hump-shaped in the product's size (45 / 111 / 47 / 22 / 6% of the linear ceiling at 0.05 / 0.075 / 0.10 / 0.15 / 0.20) while the gap is +0.019 / +0.077 / +0.020 / +0.078 / +0.022; above 0.10 the market becomes a hedge. A second nonlinear loading on x (`gvyp10x`) gives +0.049: a second interaction factor adds. `bvy25`, the Gaussian-y version of `bvych0`, has twice its gap at the same room, and the visibility ladder (`bvy25`, `bvyc0`, and `bvyc10` at `bv_comp` 1.0: +0.015 / +0.010 / +0.010) is flat within noise.

- **Two apparent failures of the mechanism are estimation cost, not structure.** `vyx` is `vyg25` at a lower price of y: its room (0.054) is about what DKKM gives up to estimation relative to the linear model (0.053), so the gap is zero, and the higher price lifts the room more than the cost. `bvysg` has twice `bvy25`'s room but DKKM's estimation loss is two and a half times larger (0.16 against 0.06): with signed exposures the nonlinear part of the tangency portfolio is a direct bet on the persistent state y, whose sample mean a 360-month window barely pins down. What matters is room net of DKKM's estimation cost. Four more GS21 wirings (investment cost, floating coupon, a larger option, state-dependent recovery) were stopped at the oracle stage with room below +0.01, and GS21's default channel is dormant at every calibration tried.

- **Knobs that are not mechanisms**: the speed of the priced state (κ 0.15 / 0.35 / 0.70 in KP14: +0.007 / +0.014 / +0.009, a mild hump within noise), the visibility of the type, the number of types (`vym3` against `vym3t3`), the sample size (`vym3` at $N = 1000$: room up, gap flat), and the ridge grid (DKKM +0.0001). The price of the interacted component is the one scaling lever (`vyx` to `vyg25`).

8 Open items

The seven defects listed on October 17 are all closed: GS21's 3a and crash-only cells are now the productivity exposure with and without the crash on one grid, BGN's group 2 cell is the quadratic price (binned to seven levels, so the multiplier reaches 1.68 rather than the cap of 3), the 41-node chain shows the discretization is not what lowers BGN's chain gaps, the reduced-form square ladder is rerun orthogonal to size, `kpgam` has ten seeds and `gvypk3` twenty.

1. The bin approximation of BGN's quadratic price: seven equiprobable bins put the outer multiplier at 1.68; more bins reach further into the tails at a cost linear in their number.

2. GS21 at the product size 0.075, where the room peaks, is the economy to build a calibration on.